

Theory of Computation

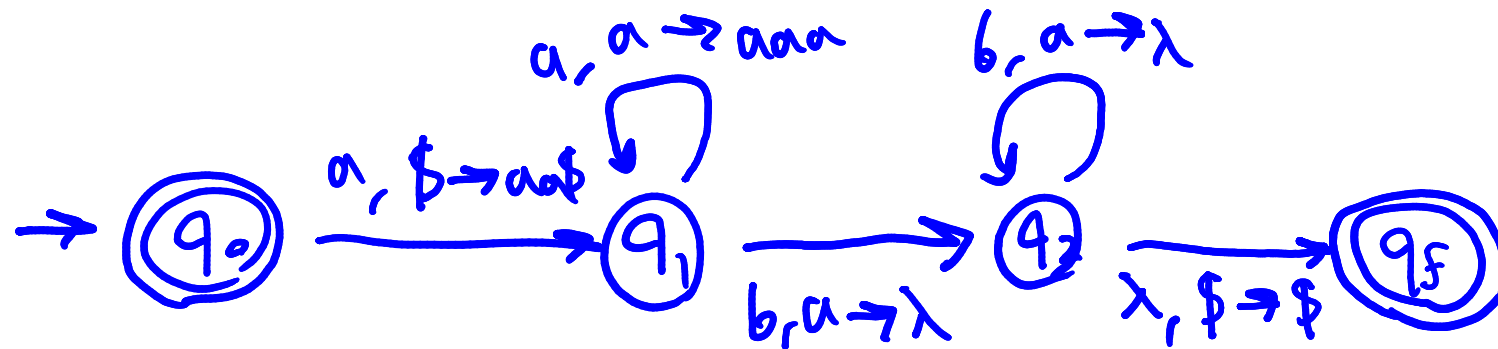
Submission link: <https://forms.gle/9QfcWiqewGmCcvG36>

Exercise 10: (Pushdown Automata)

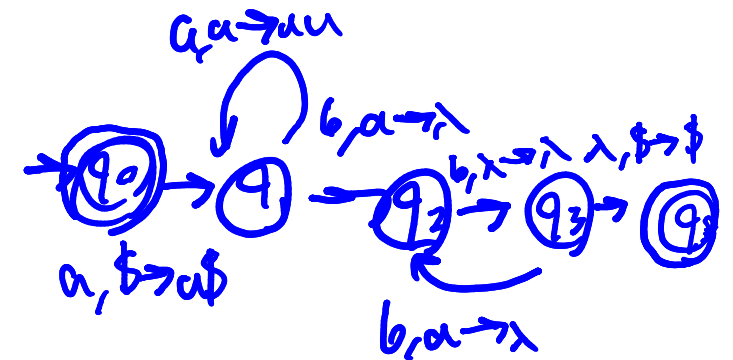
Find Pushdown Automata for the following languages.

1. $L1 = \{ a^n b^{2n} : n \geq 0 \}$

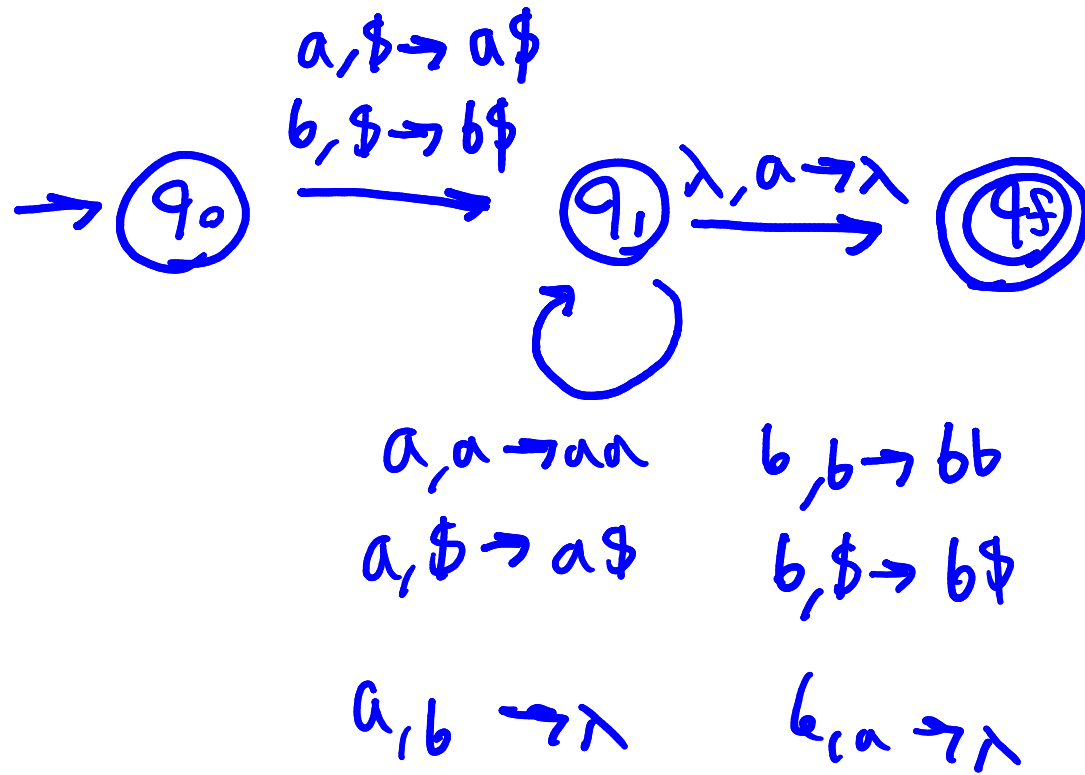
- ① If $inp. = a \rightarrow$ push a . Match b with ' a '
- ② If $inp. = a \rightarrow$ push a . Match b with ' a '



alt:



$$2. L2 = \{w \in \{a, b\}^* : n_a(w) > n_b(w)\}$$



*3. $L3 = \{a^n b^m a^{n+m} : n, m \geq 1\}$

